

EXAM1PREP

1.1 WHAT IS A GRAPH?

1.1.1. Determine which complete bipartite graphs are complete graphs.

Solution. $K_{m,n}$ is complete iff both partite sets have size at most one: $K_{1,1}$ and, under the book's convention, $K_{1,0}$ (or $K_{0,1}$). $\square$

1.1.2. Write all possible adjacency and incidence matrices for a 3-vertex path; also give adjacency matrices for P_6 and C_6 .

Solution. All arise by relabeling. Representatives for P_3 are $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ and $M = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}$. For P_6 , use the 6×6 tridiagonal matrix with ones immediately above and below the diagonal. For C_6 , additionally put ones in positions $(1, 6)$ and $(6, 1)$. $\square$

1.1.3. Using rectangular constant blocks, write an adjacency matrix for $K_{m,n}$.

Solution. $A(K_{m,n}) = \begin{pmatrix} 0 & J \\ J^T & 0 \end{pmatrix}$, with block sizes m, n . $\square$

1.1.4. Prove that $G \cong H$ iff $\overline{G} \cong \overline{H}$.

Solution. An isomorphism preserves adjacency and therefore nonadjacency, so the same bijection is an isomorphism between the complements. Apply this again for the converse. $\square$

1.1.5. Prove or disprove: if every vertex of a simple graph has degree 2, then the graph is a cycle.

Solution. False. A disjoint union of two cycles is 2-regular but is not one cycle. $\square$

1.1.6. Determine whether the pictured graph decomposes into copies of P_4 .

Solution. Yes. Its edges partition into 3-edge paths as shown in the solution-manual drawing. $\square$

1.1.7. Prove that a graph with more than six odd-degree vertices cannot be decomposed into three paths.

Solution. Every odd-degree vertex must be an endpoint of a path in a path decomposition. Three paths provide only six endpoint occurrences. $\square$

1.1.8. Show that the pictured 8-vertex graph decomposes into copies of $K_{1,3}$ and into copies of P_4 .

Solution. For the claw decomposition, use the indicated degree-3 centers and their incident edges. A second partition of the same edge set into 3-edge paths gives the P_4 decomposition displayed in the manual. $\square$

1.1.9. Prove that the graph on the right is isomorphic to the complement of the graph on the left.

Solution. Under the correspondence shown in the text, two vertices are adjacent on the right exactly when they are nonadjacent on the left. $\square$

1.1.10. Prove or disprove: the complement of a simple disconnected graph is connected.

Solution. True. Choose x, y in distinct components. Then xy is an edge of the complement. Every other vertex is adjacent in the complement to at least one of x, y ; otherwise it would connect their components in G . $\square$

1.2 PATHS, CYCLES, AND TRAILS

1.2.1. Determine whether: (a) every disconnected graph has an isolated vertex; (b) a graph is connected iff some vertex is connected to all others; (c) every closed trail's edge set partitions into cycles; (d) endpoints of a nonclosed maximal trail have odd degree.

Solution. (a) False: two disjoint edges. (b) True, by transitivity of connection. (c) True: the used-edge subgraph is even and decomposes into cycles. (d) True: an endpoint uses an odd number of incident edges; if its degree were even an unused edge would extend the trail. $\square$

1.2.2. Determine whether K_4 contains (a) a walk that is not a trail, (b) a nonclosed trail that is not a path, (c) a closed trail that is not a cycle.

Solution. (a) Yes, repeat an edge. (b) Yes, traverse a triangle and then one unused edge. (c) No nontrivial example; the length-zero closed trail at a vertex is the exception under the book's convention. $\square$

1.2.3. On $\{1, \dots, 15\}$ join i, j iff $\gcd(i, j) > 1$. Count components and find the maximum path length.

Solution. 1, 11, 13 are isolated; the remaining vertices form one component. The path 7, 14, 10, 5, 15, 3, 9, 12, 8, 6, 4, 2 spans it. Thus there are four components and maximum path length 11. $\square$

1.2.4. Describe the adjacency and incidence matrices after deleting an edge or a vertex.

Solution. Deleting an edge deletes its incidence-matrix column and removes its two symmetric adjacency entries. Deleting vertex v_i deletes row i and all incident-edge columns from the incidence matrix, and row/column i from the adjacency matrix. $\square$

1.2.5. If v is in connected G , prove that v has a neighbor in every component of $G - v$. Conclude that no cut-vertex has degree 1.

Solution. A path from a component of $G - v$ to outside it must pass through v , so v has a neighbor in that component. A cut-vertex leaves at least two components, hence has degree at least 2. $\square$

1.2.6. In the paw, find all maximal paths, maximal cliques, and maximal independent sets.

Solution. With the book's labels: maximal paths are $acb, abcd, bacd$ (last two maximum); maximal cliques are abc, cd (abc maximum); maximal independent sets are $\{c\}, \{b, d\}, \{a, d\}$ (last two maximum). $\square$

1.2.7. Prove that a bipartite graph has a unique bipartition up to swapping parts iff it is connected.

Solution. Different components may have their parts swapped independently. In a connected graph, after fixing u , parity of a path from u determines the part containing every vertex. $\square$

1.2.8. Determine when $K_{m,n}$ is Eulerian.

Solution. For $m, n > 0$, it is connected and has degrees n and m , so it is Eulerian iff both are even. If one part is empty, it has no edges and is Eulerian under the text's convention. $\square$

1.2.9. Find the minimum number of trails decomposing the Petersen graph. Can they all be paths?

Solution. All ten vertices have odd degree, so at least five trails are needed; the trail-decomposition theorem gives five. Yes: in the two-5-cycles-plus-matching drawing, use five 3-edge paths, each with one edge from each 5-cycle and one matching edge. $\square$

1.2.10. Prove or disprove: (a) every Eulerian bipartite graph has an even number of edges; (b) every Eulerian simple graph with an even number of vertices has an even number of edges.

Solution. (a) True: summing degrees over one part counts every edge once and gives an even sum. (b) False: an even cycle and an odd cycle sharing one vertex give a simple Eulerian graph of even order and odd size. $\square$

1.2.11. If Eulerian G has incident edges e, f , must some Eulerian circuit use them consecutively?

Solution. No. Take two edge-disjoint cycles sharing one vertex. Two edges at the common vertex belonging to the same cycle cannot be consecutive in an Eulerian circuit that also traverses the other cycle. $\square$

1.2.12. Convert the proof of Theorem 1.2.32 into an algorithm for finding an Eulerian circuit.

Solution. Start with a closed trail T . If unused edges remain, choose a vertex v of T incident to one. Follow unused edges from v maximally; parity forces return to v . Splice this closed trail into T and repeat. Finiteness guarantees termination with an Eulerian circuit. $\square$

1.3 VERTEX DEGREES AND COUNTING

1.3.1. Prove that a graph with exactly two odd-degree vertices contains a path joining them.

Solution. Every component has an even number of odd-degree vertices. Thus the two odd vertices lie in the same component and are joined by a path. $\square$

1.3.2. Nine students each send valentines to three others. Can every student send to exactly the people from whom they receive?

Solution. No. Symmetry would yield a 3-regular undirected graph on nine vertices, whose degree sum 27 is odd, contradicting the degree-sum formula. $\square$

1.3.3. If adjacent u, v in an n -vertex simple graph satisfy $d(u) + d(v) = n + k$, prove that uv lies in at least k triangles.

Solution. Let $S = N(u), T = N(v)$ and $j = |S \cap T|$. Then $n \geq |S \cup T| = d(u) + d(v) - j = n + k - j$, so $j \geq k$. Each common neighbor yields a triangle through uv . $\square$

1.3.4. Prove that the pictured graph is isomorphic to Q_4 .

Solution. Use the binary 4-tuple labeling shown in the solution manual. Adjacent pictured vertices differ in exactly one coordinate, precisely the adjacency rule for Q_4 . $\square$

1.3.5. Count the copies of P_3 and C_4 in Q_k .

Solution. Choose the middle vertex of P_3 and two of its k neighbors: $2^k \binom{k}{2}$. For a 4-cycle, choose the two varying coordinates and fix all others: $2^{k-2} \binom{k}{2}$. $\square$

1.3.6. Given graphs G, H , determine the number of components and maximum degree of $G + H$.

Solution. If G, H have k, ℓ components, then $G + H$ has $k + \ell$. Also $\Delta(G + H) = \max\{\Delta(G), \Delta(H)\}$. $\square$

1.3.7. Determine the maximum number of edges in a bipartite subgraph of P_n, C_n , and K_n .

Solution. P_n is bipartite, so keep all $n - 1$ edges. For C_n , keep all n edges when n is even and $n - 1$ when n is odd. For K_n , the largest bipartite subgraph is $K_{\lfloor n/2 \rfloor, \lfloor n/2 \rfloor}$ with $\lfloor n^2/4 \rfloor$ edges. $\square$

1.3.8. Decide which are graphic: (a) $(5, 5, 4, 3, 2, 2, 2, 1)$; (b) $(5, 5, 4, 4, 2, 2, 1, 1)$; (c) $(5, 5, 5, 3, 2, 2, 1, 1)$; (d) $(5, 5, 5, 4, 2, 1, 1, 1)$.

Solution. The first three are graphic; the fourth is not. Havel–Hakimi applied to (d) eventually requires a degree-3 vertex when fewer than three other positive degrees remain. $\square$

2.1 BASIC PROPERTIES

2.1.1. For each k , list the isomorphism classes of trees with maximum degree k and at most six vertices; do the same for diameter k .

Solution. For maximum degree k , begin with $K_{1,k}$ and append leaves without creating larger degree. For diameter k , begin with P_{k+1} and append leaves without creating a longer path. This generates exactly the classes in the text's table for orders at most six. $\square$

2.1.2. Let G be a graph. Prove: (a) G is a tree iff it is connected and every edge is a cut-edge; (b) G is a tree iff adding any edge with endpoints in $V(G)$ creates exactly one cycle; (c) G is a tree iff it is loopless and has exactly one spanning tree.

Solution. (a) An edge is a cut-edge iff it lies on no cycle; connected plus acyclic is a tree. (b) In a tree there is a unique path between any two vertices, so a new edge closes exactly one cycle; conversely the condition forces connectivity and forbids a pre-existing cycle. (c) A tree is its unique spanning tree. If loopless G had unique spanning tree T plus an extra edge e , then $T + e$ has a cycle and deleting another edge of that cycle gives a second spanning tree. $\square$

2.1.3. Prove or disprove: every graph with fewer edges than vertices has a component that is a tree.

Solution. True. If every component had at least as many edges as vertices, summing would give $e(G) \geq n(G)$. Hence some connected component H has $e(H) < n(H)$. Every connected graph has at least $n(H) - 1$ edges, so $e(H) = n(H) - 1$, and H is a tree. $\square$

2.1.4. Prove or disprove: every graph with fewer edges than vertices has a component that is a tree.

Solution. True, by the same component-counting argument: some component has fewer edges than vertices, and a connected graph with that property must have exactly one fewer edge than vertices and hence be a tree. $\square$

2.1.5. Prove that a maximal acyclic subgraph of G consists of a spanning tree from each component of G .

Solution. Add missing vertices as isolated vertices. In a component H , if the forest's intersection with H were disconnected, an edge of H joining two of its components could be added without creating a cycle, contradicting maximality. Thus the intersection is a spanning tree of H . $\square$

2.1.6. Let T be a tree with average degree a . Determine $n(T)$ in terms of a .

Solution. If T has n vertices, then $e(T) = n - 1$, so $a = 2(n - 1)/n$. Solving gives $n = 2/(2 - a)$. $\square$

2.1.7. Prove that every n -vertex graph with m edges has at least $m - n + 1$ cycles.

Solution. If G has k components, choose a spanning tree in each component, using $n - k$ edges. Each of the remaining $m - n + k$ edges creates a distinct cycle with the spanning forest. Since $k \geq 1$, this is at least $m - n + 1$. $\square$

2.1.8. Prove the following characterizations of forests: (a) every induced subgraph has a vertex of degree at most 1; (b) every connected subgraph is induced; (c) the number of components equals vertices minus edges.

Solution. (a) Every nontrivial tree has a leaf; conversely, a shortest cycle is induced and has all degrees 2. (b) A connected noninduced subgraph plus a missing edge between its vertices contains a cycle; conversely deleting an edge from a cycle gives a connected noninduced subgraph. (c) Sum $e(T_i) = n(T_i) - 1$ over tree components. Conversely, connected components each have at least $n_i - 1$ edges, so equality forces every component to be a tree. $\square$

2.1.9. For $2 \leq k \leq n - 1$, prove that the graph formed by adding one vertex adjacent to every vertex of P_{n-1} has a spanning tree of diameter k .

Solution. Let $v_1, \dots, v_{n-1}$ be the path and let x be universal. Use the path $v_1 \cdots v_{k-1}$ together with edges $xv_{k-1}, xv_k, \dots, xv_{n-1}$. This is a spanning tree of diameter k . $\square$

2.1.10. If u, v are vertices in a connected simple n -vertex graph and $d(u, v) > 2$, prove $d(u) + d(v) \leq n + 1 - d(u, v)$. Give examples showing failure when $d(u, v) \leq 2$.

Solution. Let $k = d(u, v)$. Since $k > 2$, $N(u) \cap N(v) = \emptyset$. On a shortest u, v path, the vertices u, v and the internal vertices except the first and last are forbidden from both neighborhoods. Thus $|N(u) \cup N(v)| \leq n + 1 - k$. For distance 2, two vertices may have many common neighbors; the degree sum can reach $2n - 4$. $\square$

2.1.11. If x, y are adjacent, prove $|d(x, z) - d(y, z)| \leq 1$ for every z .

Solution. Extend a shortest z, y path by edge yx to get $d(z, x) \leq d(z, y) + 1$; symmetrically $d(z, y) \leq d(z, x) + 1$. $\square$

2.1.12. Determine the diameter and radius of $K_{m,n}$.

Solution. If $m, n \geq 2$, both are 2. For $K_{1,n}$ with $n > 1$, radius 1 and diameter 2. For $K_{1,1}$ both are 1. For $K_{0,n}$ with $n > 1$ distances between distinct vertices are infinite; for $K_{0,1}$ both are 0. $\square$

2.1.13. Prove that every graph of diameter d has an independent set of size at least $\lceil (d + 1)/2 \rceil$.

Solution. Take a shortest path of length d between vertices at distance d . Every other vertex along this path forms an independent set, since an edge between nonconsecutive chosen vertices would shorten the path. The set has $\lceil (d + 1)/2 \rceil$ vertices. $\square$

2.1.14. Suppose computer processors are named by binary k -tuples and communicate directly iff their names are adjacent in Q_k . Given names u, v , how do you choose the first step on a shortest path from u to v ?

Solution. Their distance is the number of coordinates in which they differ. Change any one coordinate in which u and v differ; this reduces that number by one and therefore begins a shortest path. $\square$

2.1.15. Let G be simple with diameter at least 4. Prove that $\overline{G}$ has diameter at most 2.

Solution. Use the complement-diameter theorem from the text: if a simple graph has diameter at least 3, its complement has diameter at most 3, and the stronger distance-4 hypothesis rules out complement distance 3. Equivalently, any two nonadjacent vertices in $\overline{G}$ are adjacent in G and must have a common neighbor in $\overline{G}$, or G would force all routes too short. $\square$

2.1.16. Let G' be the graph on $V(G)$ in which x, y are adjacent iff $d_G(x, y) \leq 2$. Prove $\text{diam}(G') = \lceil \text{diam}(G)/2 \rceil$.

Solution. A G -path of length k yields a G' -path of length $\lceil k/2 \rceil$ by skipping every other vertex. Conversely, each edge of a G' -path replaces by a G -path of length at most 2. Hence $d_{G'}(x, y) = \lceil d_G(x, y)/2 \rceil$ for all x, y , and taking maxima gives the formula. $\square$